Digital Signal Processing

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FFT algorithm for frequency extraction based on base 2

- Principle of FFT Algorithm for Base 2 Frequency Extraction
- Flow diagram of FFT algorithm with frequency extraction of base 2

FFT algorithm for frequency extraction based on base 2

$$X[m]=\sum_{k=0}^{N-1}x[k]W_N^{mk}=\sum_{k=0}^{N/2-1}x[k]W_N^{mk}+\sum_{k=N/2}^{N-1}x[k]W_N^{mk}$$

$$N=2^M$$

$$=\sum_{k=0}^{N/2-1}x[k]W_N^{mk}+\sum_{k=0}^{N/2-1}x[k+N/2]W_N^{m(k+N/2)}$$

$$=\sum_{k=0}^{N/2-1}\left(x[k]+W_N^{mN/2}x[k+N/2]\right)W_N^{mk}$$

$$=\sum_{k=0}^{N/2-1}\left(x[k]+(-1)^m x[k+N/2]\right)W_N^{mk}$$

FFT algorithm for frequency extraction based on base 2

$$X[m]=\sum_{k=0}^{N/2-1}\left(x[k]+(-1)^m x[k+N/2]\right)W_N^{mk}$$

$$X[2m]=\sum_{k=0}^{N/2-1}\left(x[k]+x[k+N/2]\right)W_{N/2}^{mk} \qquad x_1[k] \qquad m=0,1,\cdots,\frac{N}{2}-1$$

$$X[2m+1]=\sum_{k=0}^{N/2-1}\left(x[k]-x[k+N/2]\right)W_N^{k}W_{N/2}^{mk} \qquad x_2[k] \qquad m=0,1,\cdots,\frac{N}{2}-1$$

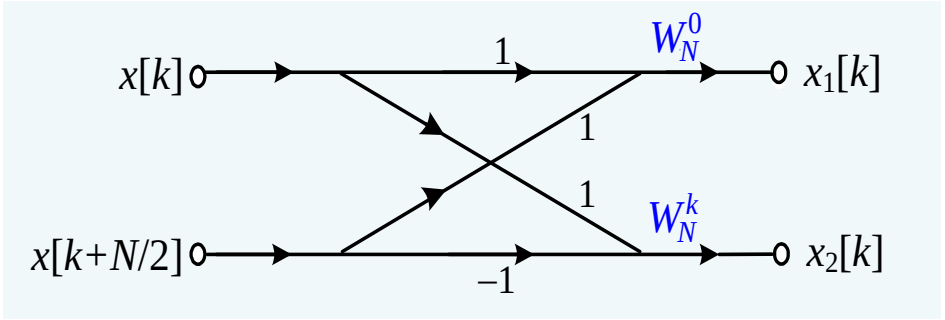
$$X[2m]=\sum_{k=0}^{N/2-1}x_1[k]W_{N/2}^{mk}$$

$$X[2m+1]=\sum_{k=0}^{N/2-1}x_2[k]W_{N/2}^{mk} \qquad m=0,1\cdots\frac{N}{2}-1$$

Decompose a long sequence into two short sequences

$$x_1[k] = x[k] + x[k + N/2]$$

$$x_2[k] = \{x[k] - x[k + N/2]\} W_N^k$$



Butterfly diagram

$$\begin{bmatrix} x_1[k] \\ x_2[k] \end{bmatrix} = \begin{bmatrix} W_N^0 & 0 \\ 0 & W_N^k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x[k] \\ x[k + N/2] \end{bmatrix}$$

FFT algorithm for frequency extraction based on base 2

Decompose the 8-point time-domain sequence into 2 4-point time-domain sequences

Each 4-point time-domain sequence is further decomposed into 2 2-point time-domain sequences

The FFT algorithm based on frequency extraction of base 2 is used to transform the time domain to the frequency domain through a 2-point sequence DFT

Time domain to frequency domain

$$\begin{bmatrix} X[0] \\ X[1] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \end{bmatrix}$$

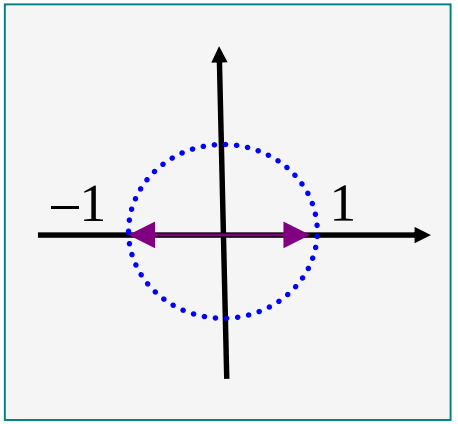
FFT algorithm for frequency extraction based on base 2

Decompose the long sequence $x[k]$ into two sets of short sequences $x_1[k]$ and $x_2[k]$

$$\begin{bmatrix} x_1[k] \\ x_2[k] \end{bmatrix} = \begin{bmatrix} W_N^0 & 0 \\ 0 & W_N^k \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x[k] \\ x[k+N/2] \end{bmatrix}$$

Time domain to frequency domain

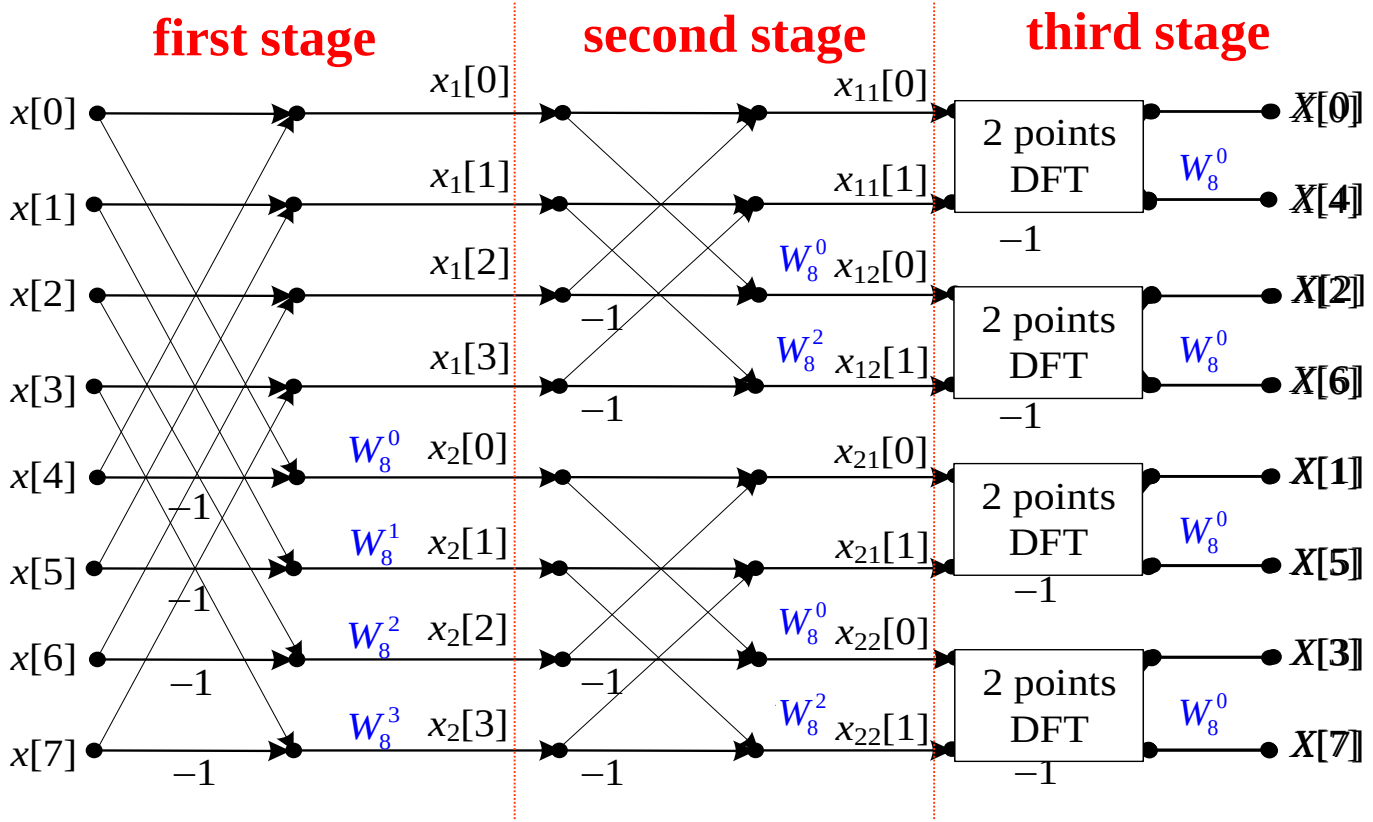
$$\begin{bmatrix} X[0] \\ X[1] \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x[0] \\ x[1] \end{bmatrix}$$



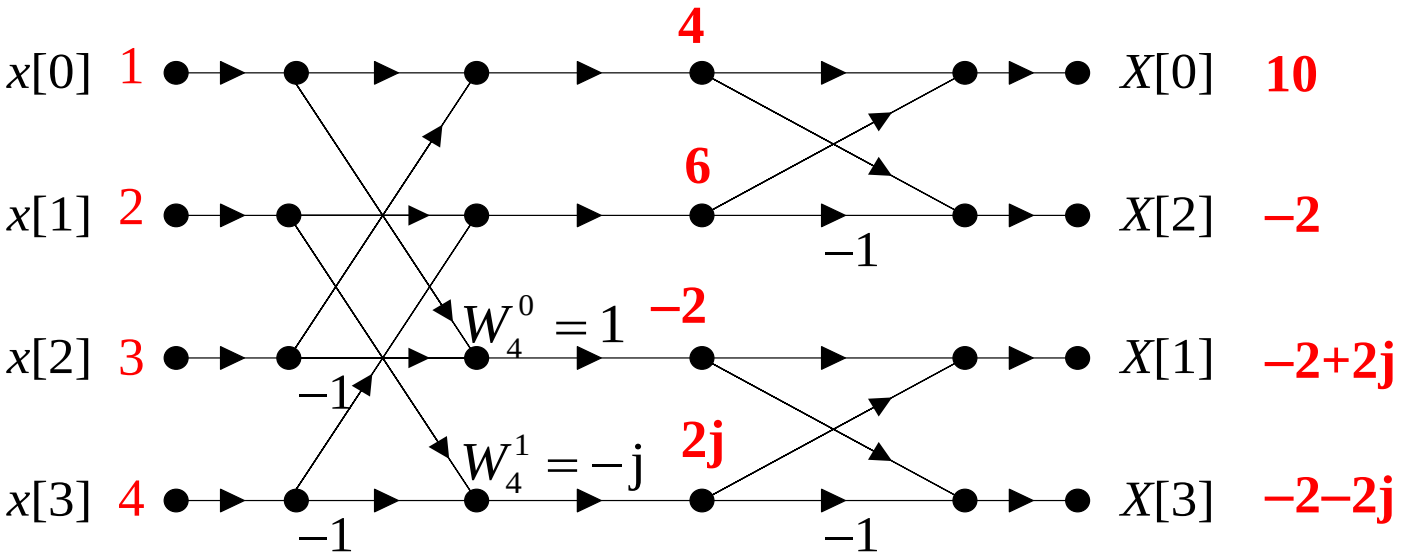
$$x_1[k] = \{x[k] + x[k+N/2]\}W_N^0$$

$$x_2[k] = \{x[k] - x[k+N/2]\}W_N^k$$

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E.g: Known $x[k]=\{1,2,3,4\}$ use the flow graph using the base 2 frequency and calculate $X[m]=\mathrm{DFT}\{x[k]\}$。



$$\mathrm{DFT}\{x[k]\}=\{10,\ -2+2\mathrm{j},\ \ -2,\ -2-2\mathrm{j}\}$$

Thanks!

The materials referenced in this course are accumulated over years of teaching by the instructor, sourced from various media and exchanges with colleagues and peers. It is difficult to indicate the source of each material one by one. This is to clarify and express gratitude!